# 10.8 - Ratio Test for Convergence

Learning Objective(s):
**LIM-7.A: Determine whether a series converges or diverges.**

## The Ratio Test

Here is another test for convergence – the **Ratio Test.**

**THEOREM 10.8-A: RATIO TEST**

Let $\sum a_n$ be a series with nonzero terms.

1. $\sum a_n$ converges if $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| < 1$.

2. $\sum a_n$ diverges if $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| > 1$ or $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \infty$.

3. The Ratio Test is inconclusive if $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = 1$.

Scan the QR Code above to watch a video covering Example 1

**Example 1: Using the Ratio Test.**
Determine the convergence or divergence of the following series.

Converges by the Ratio Test!

a. $\sum_{n=1}^{\infty}\frac{2^n}{n!} = \sum_{n=1}^{\infty} a_n$, $a_n = \frac{2^n}{n!}$

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{2^{n+1}}{(n+1)!}\cdot\frac{n!}{2^n}\right|$$

$$= \lim_{n\to\infty}\left|\frac{2}{n+1}\right| = 0 < 1$$

Converges by Ratio Test

b. $\sum_{n=1}^{\infty}\frac{n^2 2^{n+1}}{3^n} = \sum_{n=1}^{\infty} a_n$

$a_n = \frac{n^2\cdot 2^{n+1}}{3^n}$.

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(n+1)^2\cdot 2^{n+2}}{3^{n+1}}\cdot\frac{3^n}{n^2\cdot 2^{n+1}}\right| = \frac{2}{3} < 1$$

Diverges by the Ratio Test

c. $\sum_{n=1}^{\infty}\frac{n^n}{n!} = \sum_{n=1}^{\infty} a_n$, $a_n = \frac{n^n}{n!}$

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(n+1)^{n+1}}{(n+1)!}\cdot\frac{n!}{n^n}\right|$$

$$= \lim_{n\to\infty}\frac{(n+1)^n}{n^n} = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n$$

$$= e > 1$$

d. $\sum_{n=1}^{\infty}(-1)^n\frac{\sqrt{n}}{n+1} = \sum_{n=1}^{\infty}(-1)^n a_n$

$a_n = \frac{\sqrt{n}}{n+1}$

Converges by AST

$\lim_{n\to\infty} a_n = 0$

$$a_{n+1} - a_n = \frac{\sqrt{n+1}}{n+2} - \frac{\sqrt{n}}{n+1}$$

$$= \frac{(n+1)^{3/2} - \sqrt{n}(n+2)}{(n+2)(n+1)} < 0$$

# The Root Test

Although it is not tested on the AP Calculus BC exam, here is another test for convergence and divergence – the **Root Test.** You will discover that it works especially well for series involving *n*th powers.

**THEOREM 10.8-B: ROOT TEST**

Let $\sum a_n$ be a series.

**1.** $\sum a_n$ converges absolutely if $\lim_{n\to\infty} \sqrt[n]{|a_n|} < 1$.

**2.** $\sum a_n$ diverges if $\lim_{n\to\infty} \sqrt[n]{|a_n|} > 1$ or $\lim_{n\to\infty} \sqrt[n]{|a_n|} = \infty$.

**Example 2: Using the Root Test**

Determine the convergence or divergence of the following series.

$$\sum_{n=1}^{\infty} \frac{e^{2n}}{n^n} = \sum_{n=1}^{\infty} \left(\frac{e^2}{n}\right)^n$$

Converges by the Root test!

$$\lim_{n\to\infty} \sqrt[n]{\left(\frac{e^2}{n}\right)^n} = \lim_{n\to\infty} \frac{e^2}{n} = 0 < 1$$

SCAN ME

Scan the QR Code above to watch a video covering Example 2

# 10.9 - Determining Absolute or Conditional Convergence

Learning Objective(s):
**LIM-7.A: Determine whether a series converges or diverges.**

## Absolute and Conditional Convergence

The concepts of **absolute** and **conditional** convergence are used to determine the convergence (or divergence) of a series in which the terms are sometimes positive and sometimes negative – but not necessarily alternating.
We begin this analysis of series behavior with the following theorem.

**THEOREM 10.9: ABSOLUTE CONVERGENCE**

If the series $\sum |a_n|$ converges, then the series $\sum a_n$ also converges.

**Note:** The converse of the above theorem is not true. For instance, the **alternating harmonic series**

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$ converges by the Alternating Series Test. Yet the harmonic series diverges. This type of convergence is called **conditional.**

**DEFINITIONS OF ABSOLUTE AND CONDITIONAL CONVERGENCE**

1. $\sum a_n$ is **absolutely convergent** if $\sum |a_n|$ converges.
2. $\sum a_n$ is **conditionally convergent** if $\sum a_n$ converges but $\sum |a_n|$ diverges.

SCAN ME

Scan the QR Code above to watch a video covering Example 1

**Example 1: Absolute and Conditional Convergence**
Determine whether each of the series is absolutely convergent, conditionally convergent, or divergent.

**a.** $\sum_{n=0}^{\infty} \frac{(-1)^n n!}{2^n} = \frac{0!}{2^0} - \frac{1!}{2^1} + \frac{2!}{2^2} - \frac{3!}{2^3} + \cdots$

Divergent

$= \sum_{n=0}^{\infty} (-1)^n a_n, \quad a_n = \frac{n!}{2^n}$

$\lim_{n\to\infty} \frac{n!}{2^n} = \infty \neq 0$

**b.** $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}} = -\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{4}} - \cdots$

Converges Conditionally

$= \sum_{n=1}^{\infty} (-1)^n a_n, \quad a_n = \frac{1}{\sqrt{n}}$

$\lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0$

$a_{n+1} \le a_n$

$\frac{1}{\sqrt{n+1}} \le \frac{1}{\sqrt{n}}$

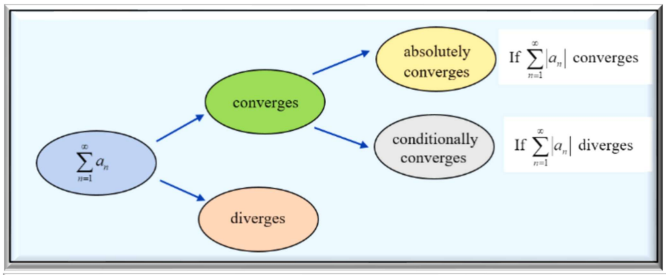
$\sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$

$\frac{1}{2} \le 1$

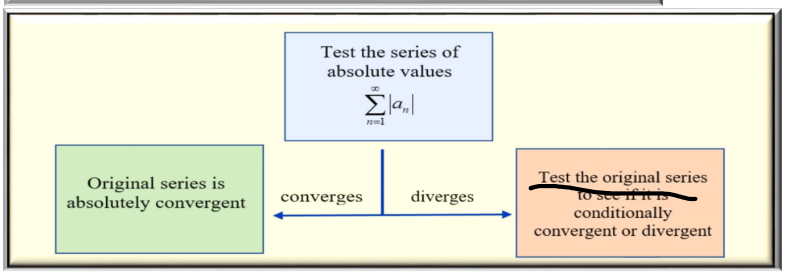
Diverges!

The following pair of flowcharts may help you when trying to determine whether a series is absolutely convergent, conditionally convergent, or divergent.

**Flowchart 1 – Absolute vs Conditional Convergence**




**Flowchart 2 – Absolute vs Conditional Convergence**




**Example 2: Absolute and Conditional Convergence**

Determine whether each of the series is absolutely convergent, conditionally convergent, or divergent.

**a.** $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}\sqrt{n}}{n^2+1}$

absolutely converges by AST

Consider $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+1}$

$\frac{1}{n^{3/2}}$

$\lim_{n\to\infty} \frac{\sqrt{n}}{n^2+1} = 0$

$f(x) = \frac{\sqrt{x}}{x^2+1}$

$f'(x) = \frac{(x^2+1)\cdot\frac{1}{2}x^{-1/2} - x^{1/2}\cdot 2x}{(x^2+1)^2} = \frac{1-3x^2}{2\sqrt{x}(x^2+1)^2} < 0$

**b.** $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}2^k}{k^2}$ Diverges

$\lim_{n\to\infty} \frac{2^k}{k^2} = \infty$

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## Rearrangement of Series

A finite sum such as $(1 + 3 - 2 + 5 - 4)$ can be rearranged without changing the value of the sum. This, however, is not necessarily true of an infinite series—it depends on whether the series is absolutely convergent
(where *every* rearrangement has the same sum) or conditionally convergent.

**Example 3: Rearrangement of a Series**

Mathematicians have shown that the alternating harmonic series $\sum_{n=1}^{\infty} (-1)^{n+1}\frac{1}{n}$ converges to $\ln 2$.

That is to say $\sum_{n=1}^{\infty} (-1)^{n+1}\frac{1}{n} = \frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \ln 2$.

Rearrange the series to produce a different sum.

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$$\frac{\pi}{4} = \left(1 + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \cdots + \frac{1}{151}\right) - \frac{1}{2} + \left(\frac{1}{153} + \frac{1}{155} + \cdots + \frac{1}{409}\right) - \frac{1}{4} + \left(\frac{1}{411} + \cdots + \frac{1}{673}\right) - \frac{1}{6} + \cdots$$